

A Gauge-Invariant Decomposition of Total Angular Momentum in QED

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MOTIVATION

Canonical decomposition (satisfies algebra, but gauge dependent):

$$\vec{J} = \vec{S}_e + \vec{L}_e + \vec{S}_\gamma + \vec{L}_\gamma$$

Physical meaning of each term:

- Electron Spin

$$\vec{S}_e = \frac{1}{2}\psi^\dagger \vec{\Sigma} \psi$$

- Electron Orbital Angular Momentum

$$\vec{L}_e = \psi^\dagger (\vec{x} \times \frac{1}{i}\nabla) \psi$$

- Photon Spin

$$\vec{S}_\gamma = \vec{E} \times \vec{A}$$

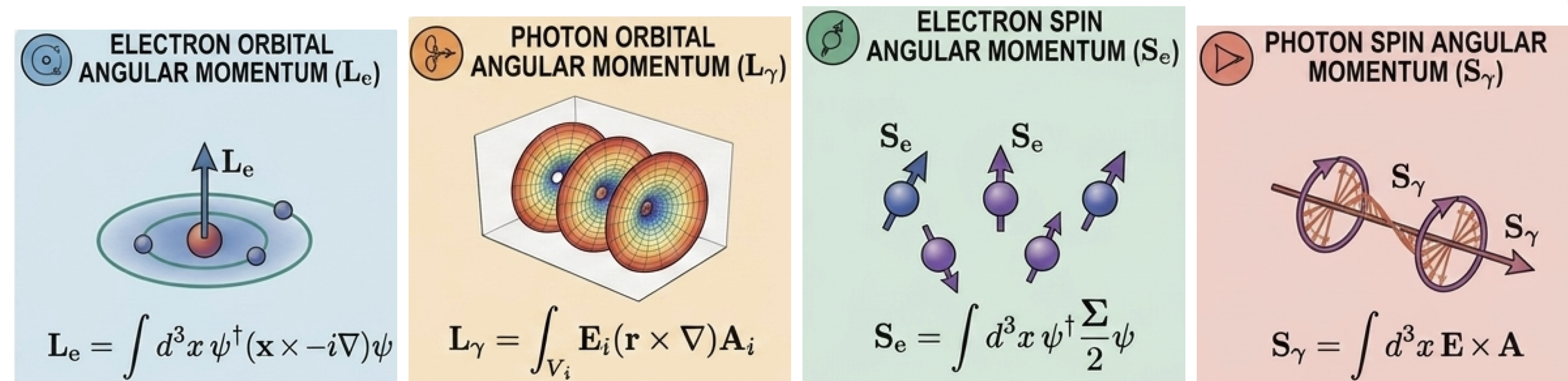
- Photon Orbital Angular Momentum

$$\vec{L}_\gamma = E^i (\vec{r} \times \nabla) A^i.$$

✓ Satisfies angular momentum algebra: $[J_i, J_j] = i\epsilon_{ijk} J_k$

× Gauge dependent – physical interpretation obscure.

We want to find a decomposition that is not only gauge invariant but also satisfies angular momentum algebra.



METHODS: Mathematical Construction

Step 1: Helmholtz decomposition of $\vec{A}$

$$\vec{A} = \vec{A}_{\text{phys}} + \vec{A}_{\text{pure}}, \quad \nabla \cdot \vec{A}_{\text{phys}} = 0, \quad \nabla \times \vec{A}_{\text{pure}} = 0.$$

Step 2: Define pure-gauge covariant derivative

$$\vec{D}_{\text{pure}} = \nabla - ie\vec{A}_{\text{pure}}$$

Step 3: Notice that $[D_i^{\text{pure}}, D_j^{\text{pure}}] = 0$

Step 4: Find that $\vec{L}_e''$ is gauge invariant

PRINCIPAL RESULTS

Physical interpretation of the Helmholtz decomposition:

- $\vec{A}_{\text{phys}}$: divergence-free – carries physical magnetic field $\vec{B} = \nabla \times \vec{A}_{\text{phys}}$
- $\vec{A}_{\text{pure}}$: curl-free – pure gauge, no physical field, can be written as $\vec{A}_{\text{pure}} = \nabla \Lambda$
- This separation uniquely separates physical degrees of freedom from gauge redundancy

Gauge-invariant decomposition:

$$\vec{J} = \vec{S}_e + \vec{L}_e'' + \vec{S}_\gamma'' + \vec{L}_\gamma''$$

use $\vec{D}_{\text{pure}}$ to substitute ∇ for electrons, use $\vec{A}_{\text{phys}}$ instead of $\vec{A}$ for photons.

Properties:

- Gauge invariant** (depends only on $\vec{A}_{\text{phys}}$ or cancels phases)
- Satisfies angular momentum algebra** $[L_e''^i, L_e''^j] = i\epsilon^{ijk} L_e''^k$
- Reduces to canonical form in Coulomb gauge ($\vec{A}_{\text{pure}} = 0$)

DISCUSSION

Belinfante decomposition:

Uses $\vec{D} = \nabla - ie\vec{A}$, gauge invariant, but fails the angular momentum algebra:

Comparison table:

	This decomposition	Belinfante	Canonical
Gauge invariant	✓	✓	No
Angular momentum algebra	✓	Only total $\vec{J}$	✓
Separates spin/orbit	✓	No	✓

This result of decomposition can be generalized in QCD, which is a Non-Abelian Gauge Theory. Their should be difference, but the idea is the same.

REFERENCES

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